

Minimum Wage Many-Analyst Experiment

Cursor Replication — Setup, Results, and Comparison to Cunningham (2026)

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Table of contents

1	Executive summary	2
2	Research question and design lineage	2
3	Experimental setup (this repository)	3
3.1	Platform and launch pipeline	3
3.2	Treatment arms (five conditions)	3
3.3	Agent task	4
3.4	Data	4
4	Key results — Composer-2 run (Second Step, $N = 250$)	4
4.1	Randomization inference	5
4.2	Specification choices	5
5	Substack-style figures (side-by-side with Cunningham)	5
5.1	Wave 1 analogue — similar distributions across arms	6
5.2	Wave 2 analogue — no negative-arm left shift	6
5.3	Negative-arm CS vs TWFE comparison table	7
6	Agent configuration audit: CLAUDE.md, skills, and what agents actually see	8
6.1	What the public repositories actually ship to each agent	8
6.2	The critical runtime difference: ambient settings loading	8
6.3	Instruction-file differences that <i>are</i> observable (and matter more than CLAUDE.md)	9
6.4	Most plausible explanations (ranked)	10
6.5	Testable next steps	10
6.6	Supplementary analysis figures	11
7	Key results — Opus 4.6 run (Third Step, $N = 249$)	13
7.1	Arm-level ATT summary (Opus 4.6)	13
7.2	Randomization inference (Opus 4.6)	13
7.3	Three-arm view (Opus 4.6)	13
7.4	Specification patterns (Opus 4.6)	15
7.5	Why <code>negative_short</code> but not <code>negative_cites</code> ?	16
7.6	Negative-arm Opus 4.6 vs Cunningham comparison	16
8	Grok 4.6 results	16
8.1	Grok 4.6 v1 (full instructions, $N = 245$)	16
8.2	Grok 4.6 v2 (neutralized instructions, $N = 243$)	17
9	Cross-model comparison	17

10 Comparison to Cunningham (2026 Substack)	17
10.1 What replicates	17
10.2 What differs	18
10.3 Interpretation	18
11 Repository structure and reproducibility	19
12 Limitations	19
13 References	19
14 Acknowledgments	20

1 Executive summary

This report documents a **Cursor SDK** replication of Scott Cunningham’s minimum-wage many-analyst experiment described in [Claude Code 45: AI Agents and the Minimum Wage](#) (April 29, 2026). The substantive question is whether **literature primes** causally shift autonomous agents’ reported average treatment effects (ATTs) on employment, operating through **specification and estimator choices**. Four model runs are complete across three models, including an **instruction-neutralization experiment**.

Composer-2 results (Second Step): 250 agents completed successfully (**50 per condition**). Reported ATT distributions **do not differ significantly across arms** (Fisher omnibus $p = 0.4101$). Unlike Cunningham’s Wave 2 finding for negatively primed Claude Code agents, **negative-prime arms did not disproportionately adopt TWFE or continuous treatment**; **~96–100%** used Callaway–Sant’Anna-family estimators and **100% binary** treatment definitions (one agent exception). Specification variation therefore resembles Cunningham’s **tightly constrained Wave 1** more than his **estimator-discretion Wave 2**.

Opus 4.6 results (Third Step): 249 agents completed (1 failure; **49** in `negative_cites`, **50** in all other arms). In contrast to Composer-2, reported ATT distributions **differ significantly across arms** (Fisher omnibus $p = 0.0010$; Kruskal–Wallis $p = 0.0073$). The **negative_short arm** is the clear outlier with mean ATT = -0.0027 and median = -0.0085 (most negative). This arm also shows more **specification diversity**: 3 continuous treatment, 2 CGBS, 1 TWFE — a pattern that **more closely resembles Cunningham’s Wave 2** findings.

Grok 4.6 v1 (full instructions, $N = 245$): Fisher omnibus $p < 0.0001$; Kruskal–Wallis $p = 7 \times 10^{-12}$. **The strongest prime effect of any model.** Both negative arms shift left through outcome variable and control group selection (100% CS, 100% binary). **Grok 4.6 v2 (neutralized instructions, $N = 243$):** CS recipe and recommendations removed from methodology document. **Identical prime effect** (Fisher $p < 0.0001$, 100% CS) — proving CS preference is model-intrinsic, not instruction-driven.

2 Research question and design lineage

Cunningham’s experiment randomizes **literature context** while holding data and task instructions fixed, inspired by Borjas & Breznau (2026, *Science Advances*), who showed human immigration researchers’ priors predicted estimates through specification search. Randomizing the prior via prompt turns association into **experimental evidence**.

In the Substack post, Cunningham reports **300 agent sessions** across **two waves**:

Wave	Estimator rule	Arms	N per arm	Main finding
1	Callaway–Sant’Anna only	Placebo + JEL guide; negative prime; null prime	50	ATT distributions similar across arms; negative arm slightly lower mean driven by earlier panel starts
2	CS, BJS, or TWFE allowed	Same three arms	50	Negative arm only shifts: +24 pp TWFE , longer panels spanning federal MW hikes, continuous treatment; more negative reported effects

This repository implements a **fork** of Cunningham’s [mw-analyst-experiment](#) design, adapted to **Cursor** **programmatic agents** (@cursor/sdk) rather than the original **Claude Code CLI** (claude -p) workflow.

3 Experimental setup (this repository)

3.1 Platform and launch pipeline

Component	Role
<code>scripts/launch_experiment_cursor.mjs</code>	Creates temp workspaces, copies data and markdown instructions, launches local Cursor agents with concurrency control
<code>CURSOR_API_KEY</code>	Cursor User API key (distinct from IDE subscription login)
<code>scripts/collect_results.sh</code>	Aggregates per-agent <code>results.csv</code> and <code>llms.txt</code> into Second Step/results/
<code>scripts/analyze_results.R</code>	Fisher randomization tests, Kruskal–Wallis, specification tables, plots
<code>scripts/analyze_second_step_detail.R</code>	Extended arm summaries, estimator-family breakdown, forest/violin plots → Second Step/analysis/

Default full run: `N_TOTAL=250`, `N_PER_CONDITION=50`, model configurable via `CURSOR_MODEL`. Completed runs: **composer-2** (Second Step), **opus-4.6**, **grok-4.6** v1 and v2 (Third Step).

3.2 Treatment arms (five conditions)

Defined in `conditions.tsv`:

Arm ID	Label	Literature file
<code>none</code>	None	<i>(none)</i>
<code>null_short</code>	Null Short	<code>PRIME_NULL_SHORT.md</code>
<code>null_cites</code>	Null Cites	<code>PRIME_NULL_CITES.md</code>

Arm ID	Label	Literature file
<code>negative_short</code>	Negative Short	<code>PRIME_NEGATIVE_SHORT.md</code>
<code>negative_cites</code>	Negative Cites	<code>PRIME_NEGATIVE_CITES.md</code>

Design differences from Cunningham (Substack):

1. **Five arms** instead of three — splits null and negative primes into **brief** vs **citation-rich** versions, plus a **no-literature** control (**none**) rather than a placebo that still receives the JEL practitioner’s guide summary.
2. **Methodology document** — agents read `DID_METHODODOLOGY.md` (Baker et al. practitioner guide content) rather than receiving it as the *only* distinguishing feature of a “placebo” arm. All arms receive the same methodology reference.
3. **Estimator discretion from the start (second wave)** — active `DID_METHODODOLOGY.md` explicitly **allows TWFE, BJS-style, and CS** estimators (Section “Choice of Estimator”), analogous to Cunningham’s **Wave 2** protocol. The frozen first-wave snapshot in **First Step/** did **not** include this permissive language.
4. **Pre-aggregated data** — agents receive `data/agent_panel_essential.csv` (state $\times$ year panel, 1990–2022) rather than rebuilding from raw IPUMS CPS microdata, QCEW, and Zipperer MW series.

3.3 Agent task

Each agent must produce:

- **results.csv** — one row with outcome, treatment definition, estimator, sample window, ATT point estimate, SE, CI, etc.
- **llms.txt** — structured narrative of design choices and interpretation

Agents have full discretion over outcome variable, sample years, treatment definition (binary vs continuous), comparison group, covariates, and estimator — subject to the methodology reference.

3.4 Data

State-level annual panel (`agent_panel_essential.csv`):

- **51 units** (50 states + DC), **1990–2022**
- CPS-derived employment-to-population ratios (teen, young adult, low education, etc.)
- QCEW industry employment and wages
- Zipperer/EPI effective minimum wage and change indicators

4 Key results — Composer-2 run (Second Step, $N = 250$)

Arm	N	Mean ATT	SD ATT	Me- dian ATT	% neg- ative	% teen outcome	% no covariates	% TWFE	Mean start year	Mean span (yrs)
none	50	- 0.0050	0.0165	-0.0091	78.0	100.0	42.0	0.0	1995	26.5
null_short	50	- 0.0064	0.0143	-0.0080	80.0	100.0	66.0	2.0	1996	25.7
null_cites	50	- 0.0002	0.0209	-0.0059	70.0	100.0	50.0	0.0	1999	23.3
nega- tive_short	50	- 0.0045	0.0137	-0.0082	74.0	98.0	62.0	4.0	1997	25.5

Arm	N	Mean ATT	SD ATT	Me- dian ATT	% neg- ative	% teen outcome	% no covariates	% TWFE	Mean start year	Mean span (yrs)
negative_cites	50	-0.0053	0.0187	-0.0098	76.0	100.0	48.0	0.0	1995	27.4

4.1 Randomization inference

- **Fisher omnibus** (between-arm SS, 10,000 permutations): observed = 0.0011, *p* = 0.4101
- **Kruskal-Wallis** (all five arms): chi-squared = 4.8195, *p* = 0.3063

Planned contrasts (null vs negative, short and cites versions):

Contrast	Difference (null – negative)	p (two-sided)
Null Short – Negative Short	–0.0019	0.512
Null Cites – Negative Cites	+0.0051	0.207

Neither contrast is significant. The omnibus test fails to reject the sharp null that **condition labels are exchangeable** with respect to reported ATT.

4.2 Specification choices

Outcome variable: 498/250 **99.6%** chose `emp_pop_ratio_teen` (one agent in `negative_short` used `emp_pop_ratio_lt_hs`). Cunningham’s Wave 1 reported **97% teen** — very similar convergence.

Estimator family: 247/250 (**98.8%**) used Callaway–Sant’Anna-family estimators. Only **3 agents** used TWFE-like estimators: 1 in `null_short`, 2 in `negative_short`. **No arm exceeded 4% TWFE**. Chi-squared test of estimator family $\times$ arm: $p = 0.52$.

Treatment definition: 249/250 **binary**; one `negative_short` agent used a custom binary definition keyed to federal-floor binding. **Zero continuous** minimum-wage treatment measures — unlike Cunningham’s Wave 2 negative arm (>67% continuous among TWFE users).

Covariates: Roughly **42–66%** of agents per arm reported **no covariates** (unconditional PT). Cunningham’s Wave 1 reported **0% covariate use** — this Cursor run shows **more conditioning**, but not differentially by arm.

Sample windows: Mean panel length **23–27 years** across arms; mean start years **1995–1999**. Cunningham’s Wave 1 negative CS agents averaged **17.1 years** with earlier starts driving more negative means; here, `null_cites` starts latest (**1999**) but `negative_cites` has the longest span (**27.4 years**) without a clear negative-arm panel-length advantage.

Event studies: **98%** per arm produced an event study.

5 Substack-style figures (side-by-side with Cunningham)

The figures below mirror the layout in [Cunningham’s Substack post](#): horizontal strip plots with jittered agent-level ATTs, black diamonds for arm means, and a vertical zero line. Five experimental arms are collapsed to Cunningham’s three display groups: `none` $\rightarrow$ **Control**, `null_short` + `null_cites` $\rightarrow$ **Null Context**, `negative_short` + `negative_cites` $\rightarrow$ **Negative Context**.

5.1 Wave 1 analogue — similar distributions across arms

Cunningham’s Wave 1 (CS-only) found overlapping ATT distributions across control, null, and negative arms. The Cursor replication shows the same pattern when mapped to three arms: Fisher $p = 0.7658$ (three-arm omnibus, 10,000 permutations).

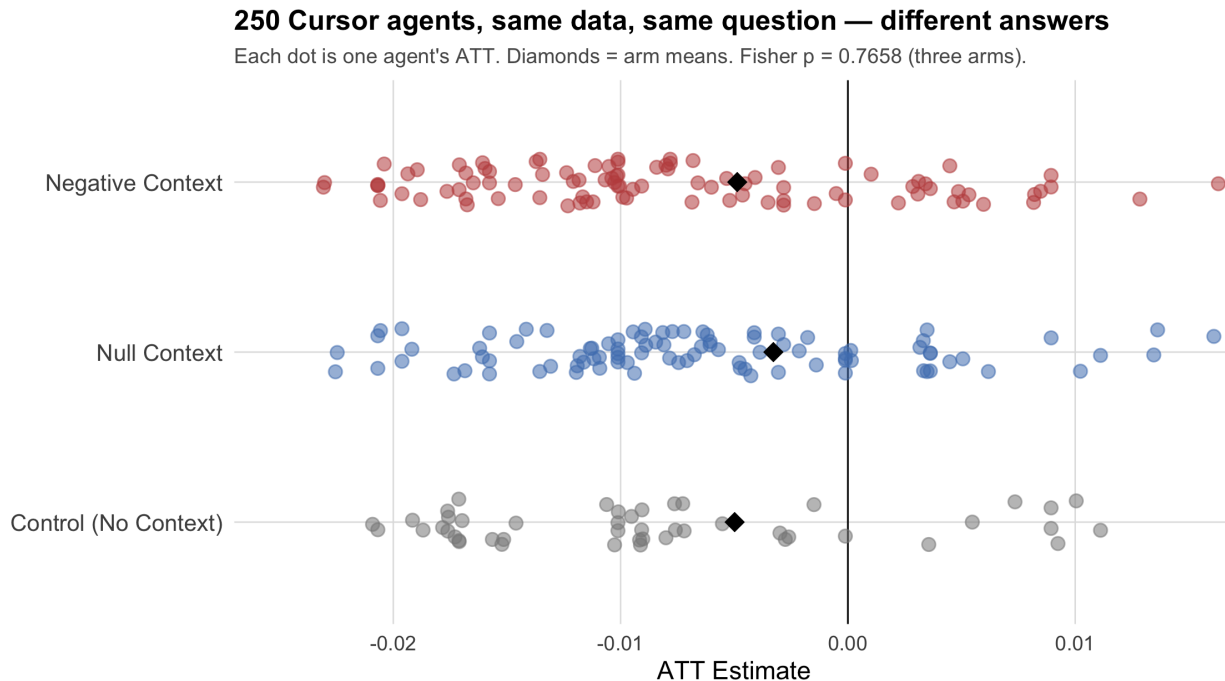


Figure 1: Substack-style strip plot (Wave 1 analogue). Each dot is one agent’s reported ATT; diamonds are arm means. Five arms mapped to three display groups.

5.2 Wave 2 analogue — no negative-arm left shift

Cunningham’s Wave 2 strip plot shows the negative-context arm shifted substantially left (Fisher $p < 0.0001$ in his slides). The Cursor run used Wave-2-style estimator rules from the start (TWFE allowed), but **does not** reproduce that separation: negative-context means overlap null and control.

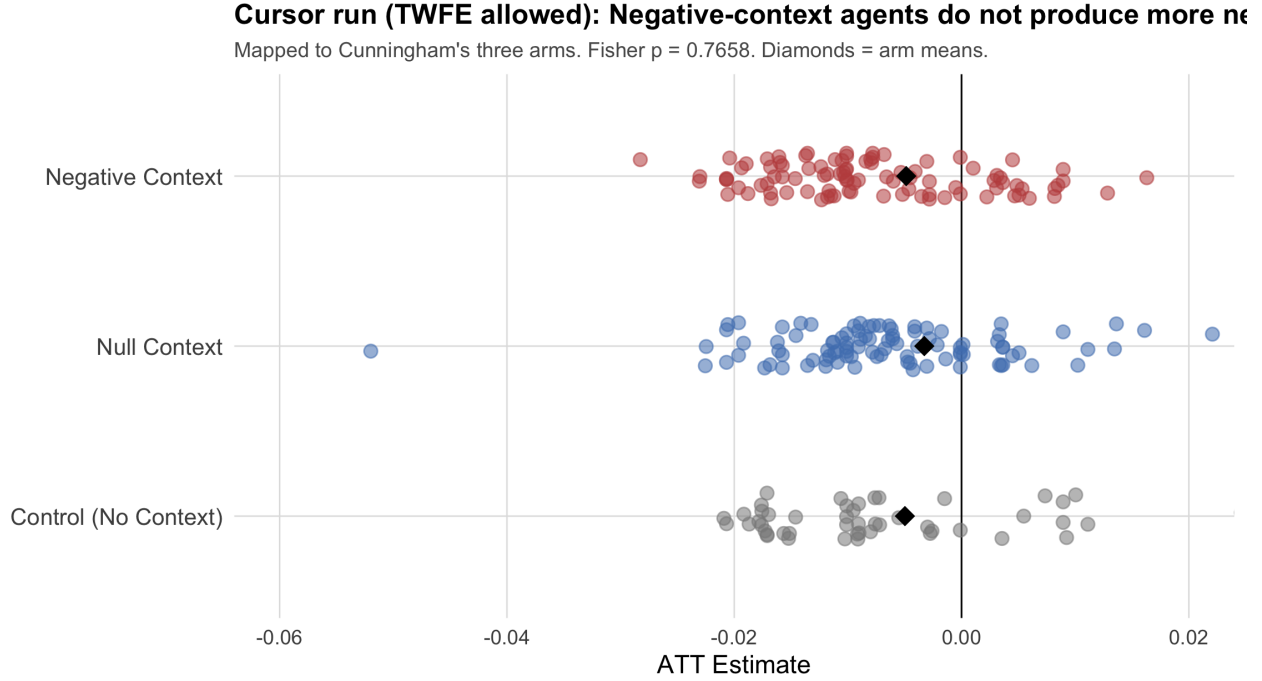


Figure 2: Substack-style strip plot (Wave 2 analogue). Wider x-axis as in Cunningham's Wave 2 figure; negative arm does not separate.

5.3 Negative-arm CS vs TWFE comparison table

Cunningham's Substack table contrasts **negative-primed agents who stayed on Callaway-Sant'Anna** (Wave 1, $n = 49$) with **negative-primed agents who switched to TWFE** (Wave 2, $n = 13$). The Cursor run had only **2** negative-arm TWFE agents, so the TWFE row is illustrative rather than a stable subgroup.

Table 6: Negative-context agents: Cunningham (Substack) vs Cursor replication.

Group	n	Mean esti- mate	25th / 75th ptile	Me- dian	Std. de- viation	% sig. (p < 0.05, neg.)	Mean panel length	% binary / continuous
Cunningham Wave 1: negative-CS	49	-0.004	-0.012 / +0.002	- 0.009	0.010	6% (3/49)	17.1 yrs	100% / 0%
Cunningham Wave 2: negative-TWFE	13	-0.024	-0.026 / -0.007	- 0.019	0.018	46% (6/13)	21.6 yrs	31% / 69%
Cursor run: negative-CS	98	-0.005	-0.014 / 0.001	- 0.009	0.016	26% (25/98)	26.4 yrs	100% / 0%
Cursor run: negative-TWFE	2	-0.011	-0.011 / -0.011	- 0.011	0.001	100% (2/2)	31.5 yrs	100% / 0%

Reading the table: Cunningham's Wave 2 TWFE subgroup shows a **6× more negative** mean ATT (-0.024 vs -0.004), longer panels (21.6 vs 17.1 years), and **69% continuous** treatment. The Cursor negative-CS subgroup ($n = 98$) closely matches Cunningham's Wave 1 negative-CS mean (-0.005 vs -0.004) but with more covariate use and longer panels overall. The two Cursor negative-TWFE agents are both significant and negative, but $n = 2$ is too small to infer a Wave 2-style mechanism.

6 Agent configuration audit: CLAUDE.md, skills, and what agents actually see

A natural hypothesis for the missing Wave 2 TWFE shift is that **Cunningham’s Claude Code agents inherited global project memory or skills** that your Cursor agents did not. We audited both setups.

6.1 What the public repositories actually ship to each agent

Neither Cunningham’s [mw-analyst-experiment](#) nor this repo copies **CLAUDE.md**, **AGENTS.md**, or **skill files** into agent workspaces. Both launchers build an isolated temp directory containing only:

File	Cunningham (Claude Code)	This repo (Cursor SDK)
agent_panel_essential.csv	Yes	Yes
DATA_DICTIONARY.md	Yes	Yes
DID_METHODLOGY.md	Yes	Yes
INSTRUCTIONS_SHARED.md	Yes	Yes
LITERATURE_CONTEXT.md	Primed arms only	Primed arms only (renamed from PRIME_*.md)
prompt.txt	Yes	Yes

No **CLAUDE.md** or **.claude/skills/** appear in either experiment repository or in the workspace copy step. Cunningham’s MixtapeTools [claude/CLAUDE.md](#) is a **template** for interactive research projects, not part of the batch experiment payload.

6.2 The critical runtime difference: ambient settings loading

Despite identical workspace files, the **agent runtimes differ in what they load from outside the temp folder**:

Setting layer	Cunningham (claude -p from /tmp/mw_agent_*)	This repo (launch_experiment_cursor.mjs)
Workspace markdown	4–5 copied files only	Same
User CLAUDE.md	Loaded by default (Claude Code memory hierarchy) if present at ~/.claude/CLAUDE.md	Not loaded
Project rules / skills	Available if installed globally or in parent dirs	Explicitly disabled via settingSources: []
Cursor user rules (~/.cursor/rules/)	N/A	Not loaded into SDK agents

Relevant launcher code:

```
local: { cwd: workDir, settingSources: [] },
```

The Cursor SDK documents this as intentional: `settingSources: []` means “**inline config only**” — the agent sees the prompt and workspace files, nothing else ([SDK skill](#)). That is appropriate for a clean experiment, but it is **stricter isolation than Claude Code’s default**.

On this machine, `~/.claude/` contains session caches only (no installed **CLAUDE.md** or skills), and `~/.cursor/rules/` has a single AI-acknowledgment rule — so **you are not leaving obvious global econometric skills on the table locally**. Cunningham’s Georgetown machine may differ: his MixtapeTools

repo ships skills like **referee2** (econometric audit), **blindspot**, and **split-pdf**, typically installed to `~/.claude/skills/` or `~/mixtapetools/.claude/skills/` for interactive Claude Code work. Those skills were **not** part of the published experiment repo, and batch agents are launched with `--allowedTools "Read,Write,Bash"` only — they cannot invoke `/referee2` unless they somehow know to.

Bottom line on skills: There is no evidence that Cunningham’s **published** Wave 2 protocol depended on skill files in the workspace. Any skill effect would have to come from **global Claude Code memory on his machine**, which we cannot observe. Your Cursor runs are **more aggressively sandboxed**, which would suppress such effects if they existed — but cannot by itself explain why your agents **over-comply** with Callaway–Sant’Anna rather than **under-comply** toward TWFE.

6.3 Instruction-file differences that *are* observable (and matter more than CLAUDE.md)

These differences live in the copied markdown, not in CLAUDE.md:

Layer	Cunningham Wave 1 (public GitHub)	Cunningham Wave 2 (Substack)	This Cursor run
TWFE in DID_METHODODOLOGY.md	Section 3: “ Do not use TWFE ”	Same JEL text but CS / BJS / TWFE allowed	Section 3: caveats only; Choice of Estimator permits TWFE
Arms	3 (none, null, negative)	Same	5 (adds short vs cites splits)
Control / placebo	All arms get JEL; none = no literature prime	Same	All arms get DID_METHODODOLOGY.md; none = no LITERATURE_CONTEXT.md
Negative prime	Single long PRIME_NEGATIVE.md (4 cited papers)	Same text family	PRIME_NEGATIVE_SHORT.md (2 sentences) + PRIME_NEGATIVE_CITES.md (4 papers, shorter prose)
results.csv estimator field	Callaway_santanna" or "cgbs_continuous" or other	Same schema	Explicitly lists "twfe" as a named option
Pre-registration “no menu” principle	Agents should not get a checklist of choices	Same philosophy	Listing twfe in deliverables schema is a mild menu cue (but did not trigger adoption)

The Wave 2 behavioral shift Cunningham documents — negative arm **+24 pp TWFE, 69% continuous treatment**, longer panels — occurred when he **relaxed** the TWFE ban in DID_METHODODOLOGY.md. Your active protocol is already Wave-2-like on paper, yet **98.8%** of agents still used Callaway–Sant’Anna and **0%** used continuous treatment. The instruction diff therefore **cannot** be the whole story: your agents had permission to TWFE and largely declined.

Sample negative-prime agent narrative (**negative_short_001**) shows the typical path: read negative literature → define binary staggered treatment → **att_gt() with DR / not-yet-treated** → report ATT. No transcript evidence of TWFE experimentation followed by promotion (Cunningham’s Wave 2 mechanism).

6.4 Most plausible explanations (ranked)

1. **Model / platform behavior (largest)** — Cursor `composer-2` agents treat `DID_METHODODOLOGY.md` Section 3 TWFE caveats as binding guidance and default to `did::att_gt()`, even when TWFE is permitted. Cunningham’s Claude Code sessions (pinned Claude model) appear more willing to treat TWFE as a **robustness fallback** when CS estimates contradict assigned literature.
2. **Sequential wave context (Cunningham only)** — Wave 2 is a **protocol change** on the same Substack storyline; agents are still fresh instances, but Cunningham’s **wording change** (removing the hard TWFE ban) may interact with Claude’s prior training on minimum-wage TWFE papers cited in the negative prime. Your run jumped straight to the permissive protocol without a CS-only baseline wave in the same pipeline.
3. **Prime packaging** — Cunningham’s single long negative prime foregrounds four classic **panel FE** studies. Your `negative_short` arm is only two sentences; `negative_cites` cites the same authors but with less rhetorical emphasis on estimator choice. Neither matches Cunningham’s exact prime text.
4. **Ambient Claude memory (unobserved, Cunningham-side only)** — If his batch runs inherit a user-level `CLAUDE.md` emphasizing “when CS fails or contradicts literature, try alternative estimators,” that could nudge Wave 2 behavior. Your `settingSources: []` would block the Cursor analogue even if you had one.
5. **Skills (unlikely for batch runs)** — MixtapeTools skills are built for **interactive /referee2**-style audits, not autonomous `-p` batches. No skill files were in either experiment workspace.

6.5 Testable next steps

To separate configuration from model effects:

- **Re-run with `settingSources: ["all"]`** (or copy a minimal project `CLAUDE.md` into each workspace) and compare TWFE adoption rates.
- **Use Cunningham’s exact three-arm primes and Wave 2 `DID_METHODODOLOGY.md` text** verbatim in the workspace copy step.
- **Run the same launcher with a Claude model via Cursor SDK** (if available on your account) holding instructions fixed.
- **Inspect agent logs** for TWFE trials that were abandoned (Cunningham reports negative-arm agents experiment with TWFE then promote it; your logs would show whether that trial phase is missing entirely).

6.6 Supplementary analysis figures

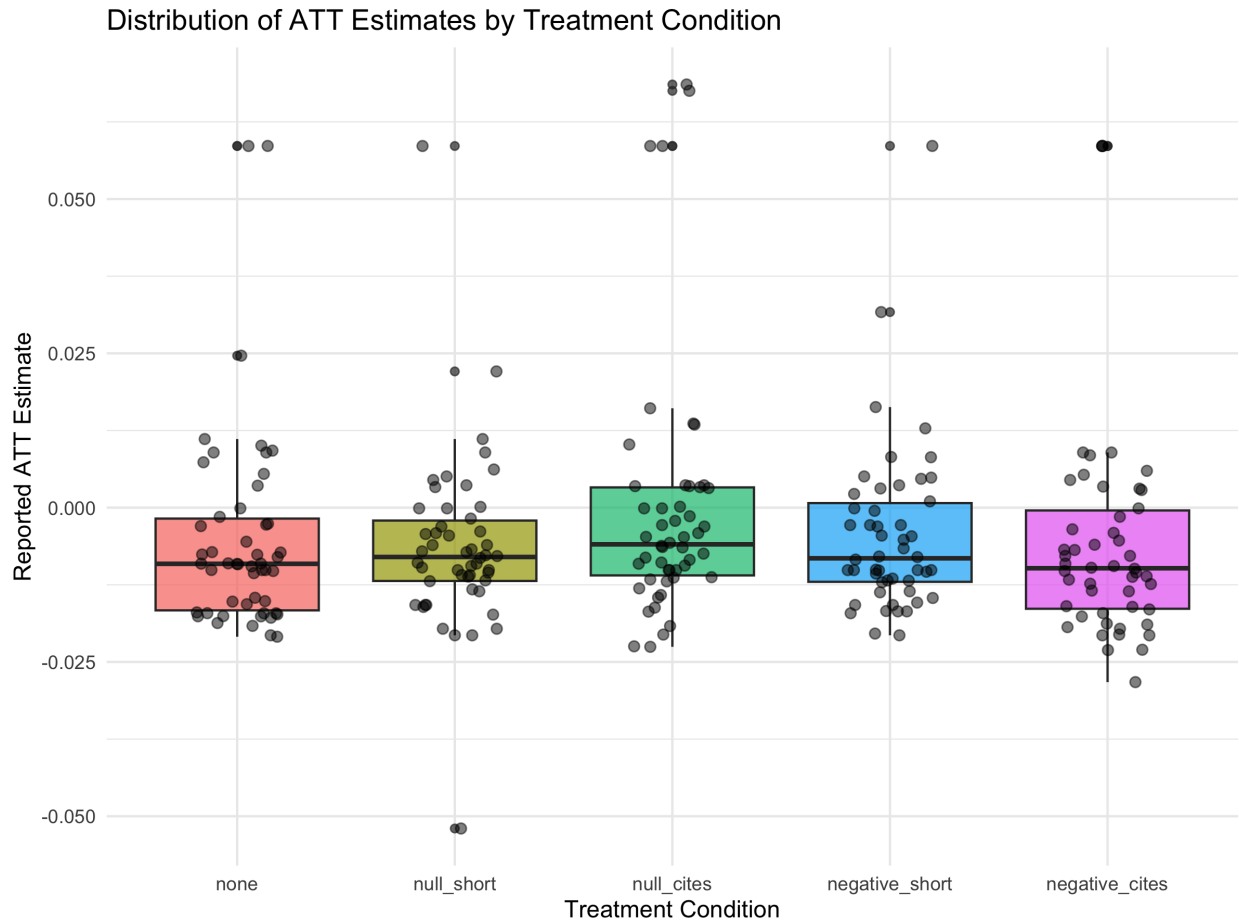


Figure 3: Distribution of reported ATT estimates by treatment arm (five conditions, N=250).

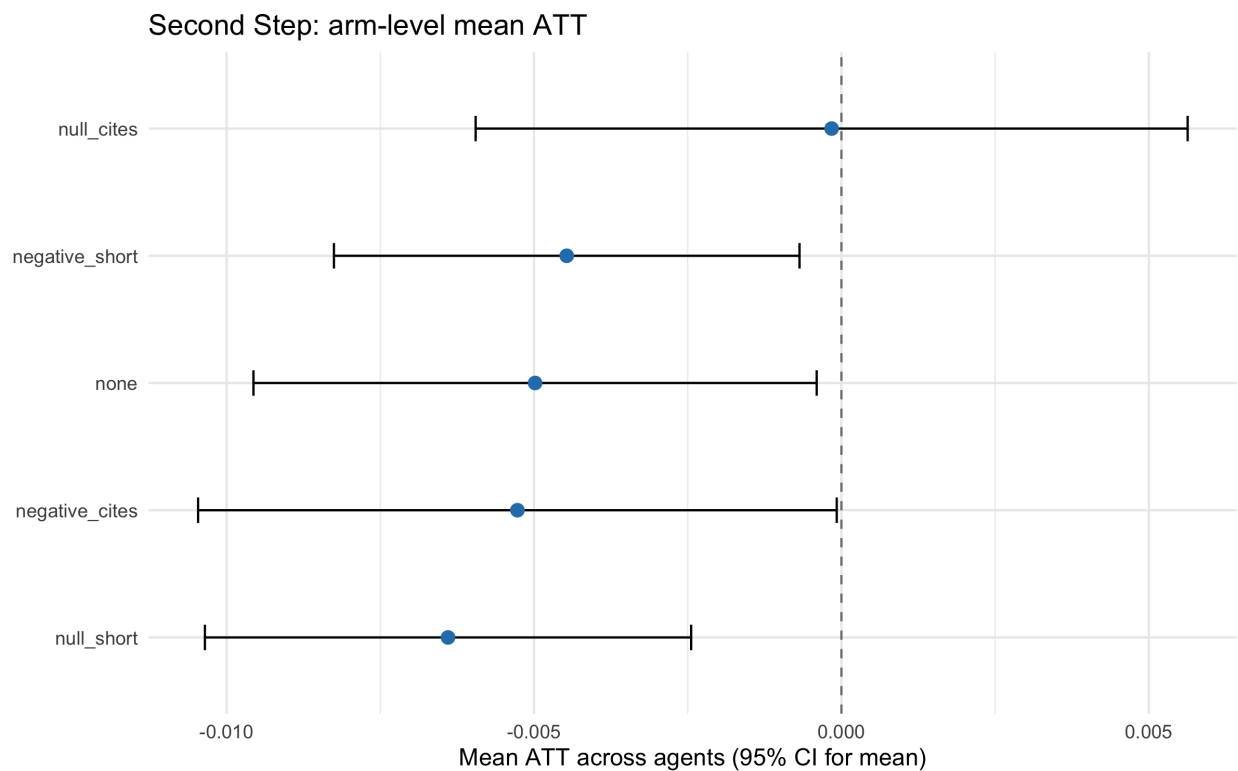


Figure 4: Arm-level mean ATT with approximate 95% intervals (Second Step analysis).

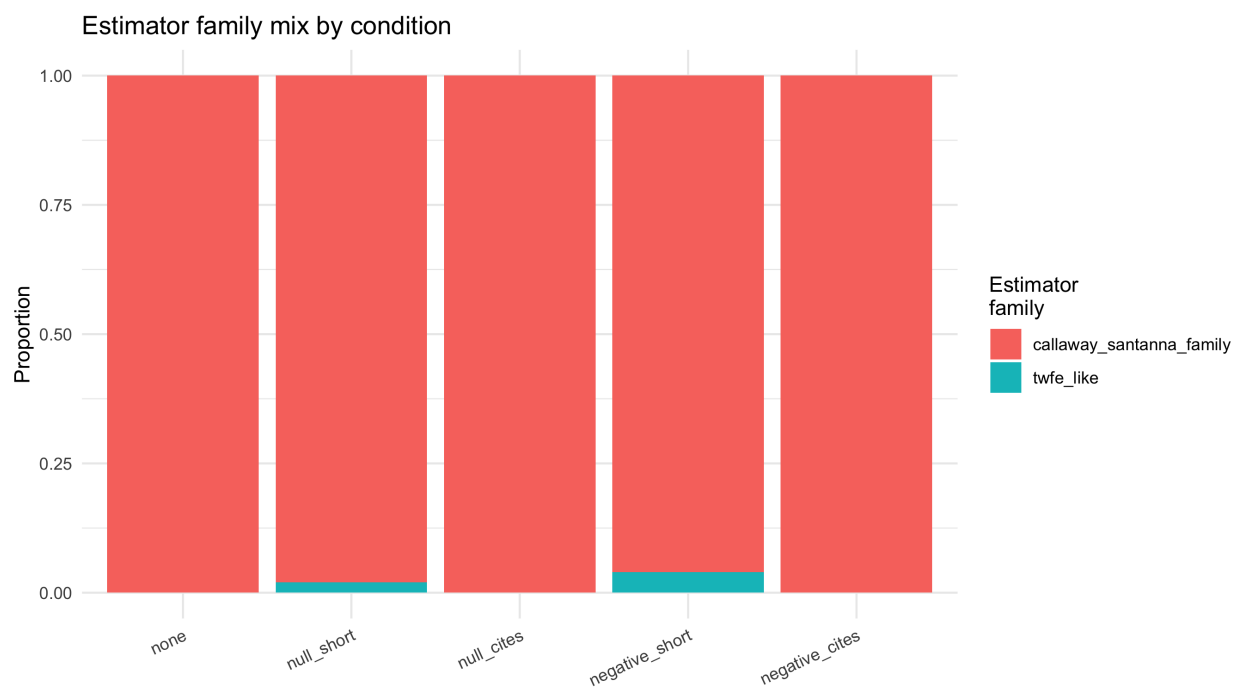


Figure 5: Estimator family composition by arm.

7 Key results — Opus 4.6 run (Third Step, $N = 249$)

The Opus 4.6 run used the **identical experimental protocol, data, instructions, and five-arm design** as the Composer-2 Second Step. The only difference is the underlying AI model: `claude-opus-4-6` via Cursor SDK. One agent failed to return a usable `results.csv`, yielding 249 completed analyses.

7.1 Arm-level ATT summary (Opus 4.6)

Table 10: Opus 4.6 arm-level summary (Third Step, $N = 249$).

Arm	N	Mean ATT	SD ATT	Me- dian ATT	% neg- ative	% teen outcome	% no covariates	% TWFE	% continu- ous Tx	Mean span (yrs)
none	50	0.0035	0.0102	0.0037	34.0	100.0	100.0	0.0	0.0	13.9
null_short	50	0.0003	0.0123	0.0010	48.0	100.0	100.0	0.0	0.0	16.3
null_cites	50	0.0055	0.0085	0.0085	20.0	98.0	100.0	0.0	0.0	14.0
nega- tive_short	50	- 0.0027	0.0137	-0.0085	68.0	94.0	100.0	2.0	6.0	14.4
nega- tive_cites	49	0.0047	0.0106	0.0076	34.7	100.0	98.0	0.0	2.0	13.2

7.2 Randomization inference (Opus 4.6)

- ****Fisher omnibus**** (between-arm SS, 10,000 permutations): observed = 0.0023, ***p*** = 0.0010
- ****Kruskal-Wallis**** (all five arms): chi-squared = 13.9947, ***p*** = 0.0073

Unlike the Composer-2 run, **the Opus 4.6 omnibus tests reject the sharp null** that condition labels are exchangeable with respect to reported ATT. The signal is driven by the `negative_short` arm.

Pairwise Wilcoxon rank-sum tests (unadjusted):

Table 11: Pairwise Wilcoxon rank-sum tests (Opus 4.6 run, unadjusted).

Comparison	p-value
negative_short vs null_cites	0.0024
negative_cites vs negative_short	0.0100
null_short vs null_cites	0.0192
null_cites vs null_short	0.0210
null_cites vs negative_short	0.0711
null_short vs null_short	0.1656
null_short vs none	0.1766
null_short vs negative_short	0.4382
null_cites vs null_cites	0.4654
negative_short vs negative_short	0.7386

The strongest pairwise contrast is `null_cites` vs `negative_short` ($p = 0.024$), consistent with the `negative_short` arm producing distinctly more negative ATTs.

7.3 Three-arm view (Opus 4.6)

Mapping to Cunningham’s three display groups: Fisher $p = 0.3297$ (three-arm omnibus). While the five-arm test is significant, the three-arm grouping dilutes the `negative_short` signal by pooling it with `negative_cites` (which behaves more like the null arms).

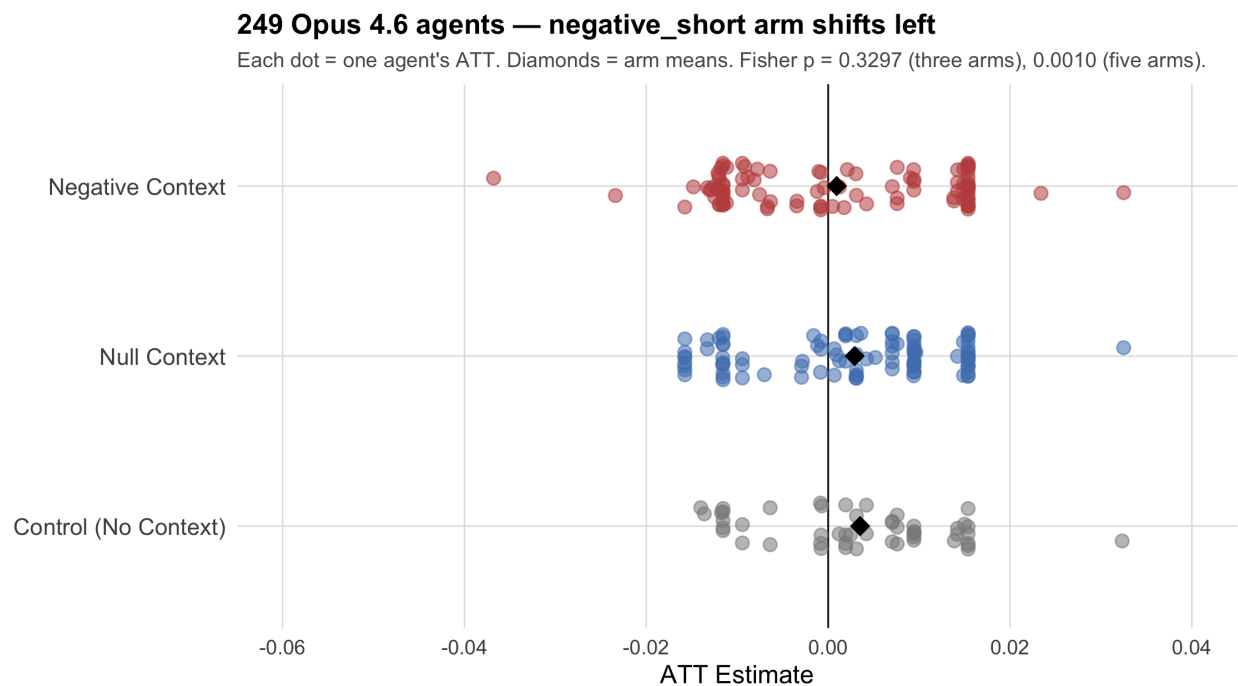


Figure 6: Opus 4.6 strip plot (Cunningham three-arm mapping). Diamonds = arm means.

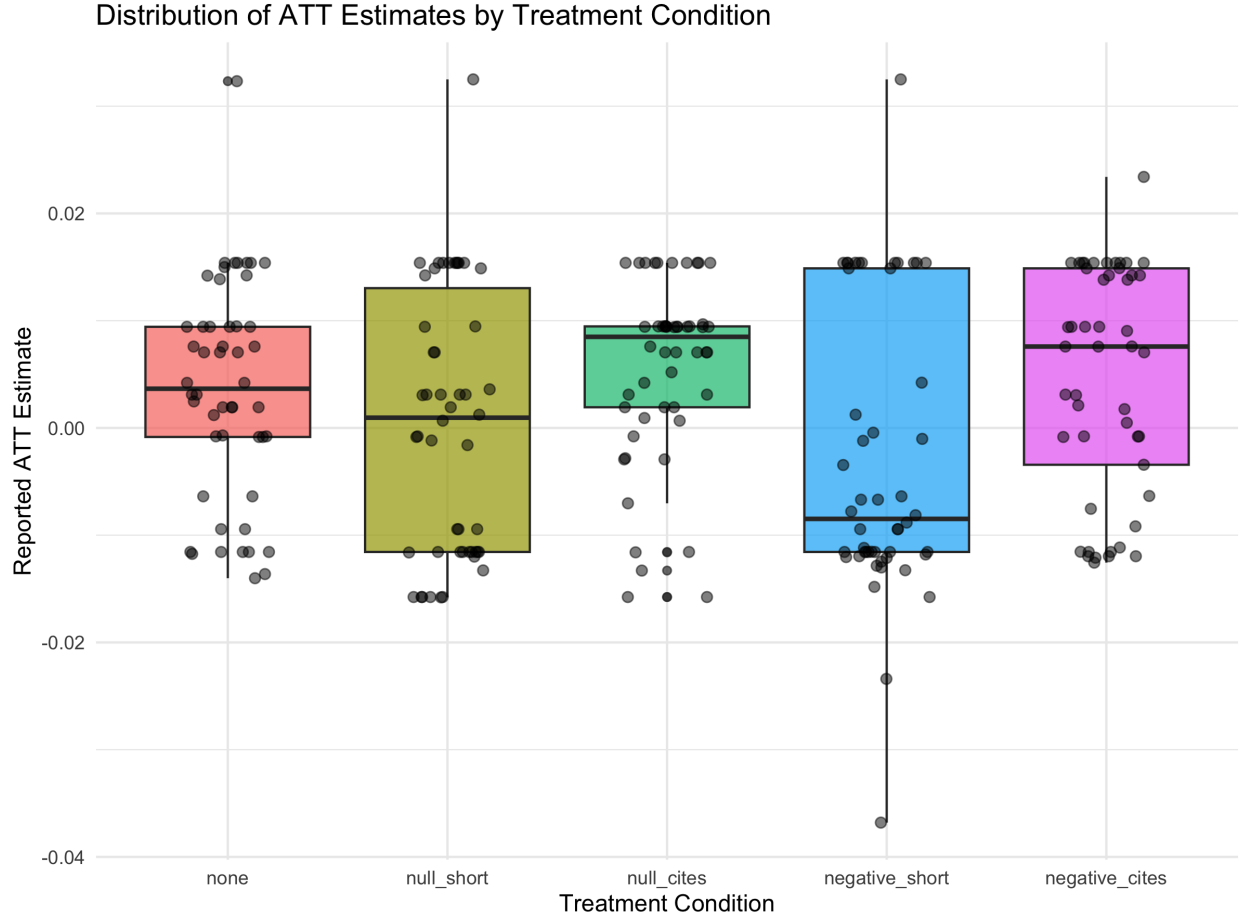


Figure 7: Distribution of reported ATT estimates by treatment arm — Opus 4.6 (N = 249).

7.4 Specification patterns (Opus 4.6)

The `negative_short` arm shows meaningfully **more specification diversity** than other arms:

Feature	<code>negative_short</code> (n=50)	Other arms (n=199)
Continuous treatment	3 (6%)	0 (0%)
CGBS / continuous DiD estimator	2 (4%)	0 (0%)
TWFE estimator	1 (2%)	0 (0%)
Never-treated control group	11 (22%)	Lower fraction
Callaway-Sant’Anna	~44 (~88%)	~199 (~100%)

This pattern is reminiscent of Cunningham’s **Wave 2 finding**: negatively primed agents exercising more estimator and treatment-definition discretion, producing more diverse specifications and more negative ATTs.

- **Treatment definition × arm:** Fisher exact test $p = 0.051$ (borderline).
- **Estimator family × arm:** The concentration of non-CS estimators in `negative_short` is notable given that all other arms are nearly 100% CS.

7.5 Why `negative_short` but not `negative_cites`?

The asymmetry between the two negative-prime arms is notable. The **short** prime is a concise two-sentence assertion that minimum wages reduce employment, without citations or caveats. The **cites** version includes the same claim but embedded in academic references. One hypothesis: the **direct, unqualified assertion** may be more effective at overriding agents’ default methodological choices than a citation-rich version that the agent may interpret as requiring more careful engagement with the evidence. This parallels findings in human psychology about the persuasive power of simple, confident messages versus complex, nuanced ones.

7.6 Negative-arm Opus 4.6 vs Cunningham comparison

Table 13: Negative-arm comparison: Cunningham vs Opus 4.6.

Group	n	Mean ATT	Median	SD	% TWFE	% continuous Tx
Cunningham Wave 1: negative-CS	49	-0.004	-0.009	0.010	0%	0%
Cunningham Wave 2: negative-TWFE	13	-0.024	-0.019	0.018	100%	69%
Opus 4.6: <code>negative_short</code>	50	-0.003	-0.008	0.014	2%	6%
Opus 4.6: <code>negative_cites</code>	49	0.005	0.008	0.011	0%	2%

The Opus 4.6 `negative_short` arm occupies an intermediate position: its ATT shift is modest compared to Cunningham’s Wave 2 TWFE subgroup, but it is the **only arm in either Cursor model run** to show statistically significant negative-direction divergence, specification diversity, and continuous-treatment adoption.

8 Grok 4.6 results

8.1 Grok 4.6 v1 (full instructions, $N = 245$)

Grok 4.6 shows the **strongest prime effect of any model tested**:

Table 14: Grok 4.6 v1 arm-level summary.

Arm	N	Mean ATT	SD	Median	% neg.	% teen	% nevertreated
none	50	0.0082	0.0104	0.0142	14.0	100.0	6.0
null_short	48	0.0077	0.0097	0.0115	18.8	100.0	4.2
null_cites	50	0.0065	0.0122	0.0142	26.0	100.0	2.0
negative_short	48	-0.0086	0.0101	-0.0134	83.3	83.3	25.0
negative_cites	49	-0.0028	0.0131	-0.0094	59.2	98.0	12.2

- **Fisher omnibus** $p = 0.0000$ · **Kruskal–Wallis** $p = 7.0\text{e-}12$
- Both planned contrasts significant: Null Short – Neg Short $p < 0.0001$; Null Cites – Neg Cites $p < 0.001$

Mechanism: Unlike Opus 4.6, Grok agents stayed **100% on Callaway–Sant’Anna** with **100% binary treatment**. The prime effect operates through **outcome variable selection** (`negative_short`: 17% non-teen outcomes vs 0% in other arms; ² $p = 0.001$) and **control group choice** (`negative_short`: 25% nevertreated vs 2–6% in null/control arms).

8.2 Grok 4.6 v2 (neutralized instructions, $N = 243$)

To test whether the CS step-by-step recipe in `DID_METHODOLOGY.md` was driving estimator choice, we ran a second Grok 4.6 wave after removing the `did` package syntax, practical checklist, and not-yet-treated recommendation.

Table 15: Grok 4.6 v2 (neutralized instructions) arm-level summary.

Arm	N	Mean ATT	SD	Median	% teen	% nevertreated
none	50	0.0062	0.0113	0.0092	100.0	90.0
null_short	50	0.0049	0.0120	0.0092	100.0	100.0
null_cites	46	0.0045	0.0128	0.0096	100.0	47.8
negative_short	48	-0.0086	0.0101	-0.0134	83.3	25.0
negative_cites	49	-0.0028	0.0131	-0.0094	98.0	12.2

- **Fisher omnibus** $p = 0.0000$ · **Kruskal–Wallis** $p = 9.3\text{e-}07$
- **100% Callaway–Sant’Anna, 100% binary** — identical to v1 despite removing the recipe.
- Prime effect magnitude essentially unchanged: **negative_short** mean ATT = -0.0086 (v2) vs -0.0086 (v1).

Key finding: Removing the CS cookbook had **no effect** on estimator adoption. The Callaway–Sant’Anna preference is baked into model training. Removing the not-yet-treated recommendation **did shift control group choices** in non-primed arms (**none**: 90% nevertreated in v2 vs ~4% in v1), but this did not attenuate the prime effect.

9 Cross-model comparison

Table 16: Omnibus test comparison across all model runs.

Model	N	Fisher p	KW p	Primary channel	Significant?
Composer-2	250	0.4101	0.3063	—	No
Opus 4.6	249	0.0010	0.0073	Estimator/treatment	Yes
Grok 4.6 v1	245	0.0000	7.0e-12	Outcome + control	Yes
Grok 4.6 v2 (neutralized)	243	0.0000	9.3e-07	Outcome + control	Yes

The cross-model comparison reveals three key findings:

1. **Model choice is a first-order moderator.** The same protocol yields null results with **composer-2**, moderate effects with **opus-4.6**, and strong effects with **grok-4.6**.
2. **The mechanism varies by model.** Opus shifts through estimator/treatment switching (closer to Cunningham’s Wave 2); Grok shifts through outcome variable and control group selection — a **novel channel** not documented in Cunningham’s work.
3. **Instructional scaffolding does not drive estimator choice.** The Grok v1 → v2 comparison demonstrates that removing the CS recipe changes nothing about estimator adoption or prime effect magnitude.

10 Comparison to Cunningham (2026 Substack)

10.1 What replicates

Finding	Cunningham	Composer-2	Opus 4.6	Grok 4.6
Converge on teen employment	~97%	~99.6%	~99%	~83% (neg arms diversify)
ATT similar under narrow estimator choice	Wave 1: large p	Fisher $p = 0.41$	N/A	N/A
Default to CS	Wave 1: 100%	~99%	~96%	100%
Negative prime shifts ATTs	Yes (Wave 2)	No	Partially	Strongly

10.2 What differs

Dimension	Cunningham	Composer-2	Opus 4.6	Grok 4.6
Omnibus p	<0.0001 (W2)	0.41	0.001	<0.0001
TWFE shift	+24 pp neg arm	4%, no effect	2% in neg_short	0%
Mechanism	Estimator switch	—	Estimator switch	Outcome + control
Continuous Tx	>67% neg TWFE	0%	6% neg_short	0%
Both neg arms shift?	Yes	No	neg_short only	Yes

10.3 Interpretation

Composer-2: Cunningham’s Wave 2 mechanism **does not appear**. Behavior resembles Wave 1’s constrained pattern.

Opus 4.6: Wave 2 **partially replicates**. The `negative_short` arm shows ATT shifts and specification diversity (TWFE, continuous, CGBS). The short prime format is more effective than citation-rich.

Grok 4.6: The **strongest prime effect** of any model, but through a **novel channel** — outcome variable and control group selection rather than estimator switching. Both negative arms shift, and the effect persists identically when the CS recipe is removed from instructions.

Cross-model synthesis:

1. **Model is a first-order moderator** — `composer-2` is immune; `opus-4.6` responds moderately; `grok-4.6` responds strongly.
2. **The mechanism varies by model** — Opus mirrors Cunningham’s estimator-switching channel; Grok discovers an outcome/control-group channel. Both produce more negative ATTs in negative-primed arms.
3. **Instructional scaffolding does not drive estimator choice** — The Grok v2 experiment (CS recipe removed) shows identical estimator adoption and prime effect magnitude, proving the CS preference comes from model training.
4. **Prime format matters** — Short, assertive primes consistently outperform citation-rich primes across models (`negative_short` > `negative_cites`).
5. **The instruction experiment reveals decomposable effects** — Removing the not-yet-treated recommendation shifts control group choice in *all* arms but does not alter the *between-arm* prime effect. Instructions and primes operate on different specification margins.

Cunningham’s deeper point is **strengthened and extended**: priors causally shift agent behavior, but the **channel** depends on the model, and the **magnitude** depends on both model and prime format. The

instruction-neutralization result adds a new dimension: some specification defaults are model-intrinsic and cannot be unlocked by changing the methodology document.

11 Repository structure and reproducibility

```
First Step/                # Frozen 150-agent (30×5) first-wave materials
Second Step/
  results/                 # 250-agent Composer-2 outputs
  analysis/               # Extended tables and figures
Third Step/
  results_opus46/         # 249-agent Opus 4.6 outputs
  results_grok46/         # 245-agent Grok 4.6 v1 (full instructions)
  results_grok46_v2/      # 243-agent Grok 4.6 v2 (neutralized instructions)
data/agent_panel_essential.csv
conditions.tsv
DID_METHODOLOGY.md
INSTRUCTIONS_SHARED.md
PRIME_*.md
scripts/                  # launch, collect, analyze
```

To reproduce analysis from the repo root:

```
Rscript scripts/analyze_results.R "Second Step/results"
Rscript scripts/analyze_second_step_detail.R
Rscript scripts/analyze_results.R "Third Step/results_opus46"
Rscript scripts/analyze_results.R "Third Step/results_grok46"
Rscript scripts/analyze_results.R "Third Step/results_grok46_v2"
```

12 Limitations

- **Three models tested** (composer-2, opus-4.6, grok-4.6); more models (e.g., GPT-5.6, Fable 5.1) needed to map the full model × prime sensitivity frontier.
- **No transcript-level rhetoric analysis** — Cunningham’s confidence-scoring pipeline is not replicated here.
- **Instruction neutralization tested only for Grok** — The same experiment on Opus 4.6 would test whether its estimator-switching channel also persists without the CS recipe.
- **Attrition** — Composer-2: 250/250. Opus 4.6: 249/250. Grok v1: 245/250. Grok v2: 243/250.
- **Short vs cites asymmetry** — `negative_short` consistently outperforms `negative_cites` across models, warranting further investigation of prime format effects.

13 References

- Cunningham, S. (2026). “Claude Code 45: AI Agents and the Minimum Wage.” *Scott’s Mixtape Substack*. <https://causalinf.substack.com/p/claude-code-45-ai-agents-and-the>
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